

Pure states and faithfulness

[Back to Project mathematical routes](#)

Scope

A pure state detects a hypothetical nonzero kernel element and its GNS representation is irreducible. Unitary equivalence to the singleton representative then contradicts that kernel element, giving faithfulness and the related simplicity consequence.

5 direct Cards + 5 reused prerequisites = 10 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [Faithfulness from a unique irreducible class](#)
- Key result: [A faithful singleton model forces simplicity](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#)

Level 0 (2 Cards)

Level 0

Pure states as real extreme points

Defines purity inside the normalized positive state space.

`MathlibAnnex.Analysis.CStarAlgebra.IsPureState`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The GNS representation of a pure state is irreducible](#) · [A character extends to a pure state](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Representations without a unit-preservation requirement

Names $*$ -representations of a complex C^* -algebra on a Hilbert space when no unit equation is required.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (4 Cards)

Level 1

The GNS representation of a pure state is irreducible

Applies the cyclic pure-state criterion to the space and vector constructed from the same state.

`MathlibAnnex.Analysis.CStarAlgebra.isIrreducible_pureState_gnsStarAlgHom`

Immediate Card prerequisites: [Pure states as real extreme points](#)

Used by in this scope: [A faithful singleton model forces simplicity](#) · [Faithfulness from a unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A character extends to a pure state

Chooses an extreme state in the compact face of all extensions of one prescribed character.

`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_extension`

Immediate Card prerequisites: [Pure states as real extreme points](#)

Used by in this scope: [A pure state detects a nonzero square](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Unitary equivalence of two representations

Defines equivalence by one surjective isometry intertwining all algebra actions.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.UnitaryEquivalent`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Nonzero irreducibility without a unit assumption

Requires nonzero action as well as the absence of proper closed reducing subspaces.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsIrreducible`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2 (3 Cards)

Level 2

A pure state detects a nonzero square

Detects a nonzero element through its positive square without an algebra separability assumption.

`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_nonzero_on_star_mul_self`

Immediate Card prerequisites: [A character extends to a pure state](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A faithful singleton model forces simplicity

Uses irreducible representations annihilating proper ideals to rule out nonzero proper closed ideals.

`MathlibAnnex.Analysis.CStarAlgebra.Representation.isSimpleCStarAlgebra_of_singleton_of_injective`

Immediate Card prerequisites: [The GNS representation of a pure state is irreducible](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 2

A representative of the unique irreducible class

Defines when one nonzero irreducible $*$ -representation represents the sole unitary-equivalence class of such representations.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsSingletonIrreducibleModel`

Immediate Card prerequisites: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (1 Card)

Level 3

Faithfulness from a unique irreducible class

Uses a pure state on the unitization to detect any hypothetical nonzero element of the kernel.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.injective_of_singleton`

Immediate Card prerequisites: [A representative of the unique irreducible class](#) · [A pure state detects a nonzero square](#) · [The GNS representation of a pure state is irreducible](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)